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Palaeomagnetic data from central Mexico and their use for palaeosecular variation studies

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ABSTRACT

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Three Brunhes-age palaeomagnetic data sets from central Mexico exhibit different dispersion of virtual geomagnetic poles. Volcanics from the eastern part of the Trans-Mexican Volcanic Belt result in a $S_F = 14.8°$. This value is interpreted to represent an upper limit, as these rocks probably are affected by localized tectonic rotations. For a stalagmite, $S_F = 9.5°$, which is interpreted as a lower limit, as the stalagmite is only 1200 years old and the data are strongly biased in time. Seventy-four volcanic units from the Chichinautzin Formation result in $S_F = 11.7°$, with confidence limits of 10.4° and 13.3°. This dispersion is much lower than expected from palaeosecular variation models, but corresponds to data obtained for other areas in or beneath the Pacific Basin, e.g. the Hawaiian islands.

1. Introduction

The study of the palaeosecular variation (PSV) of the Earth's magnetic field was one of the first applications of palaeomagnetic data. Pioneer studies of lava flow sequences from several parts of the world (e.g. Cox, 1962; Creer, 1962a,b) permitted estimation of palaeosecular variation (PSV) for periods beyond observatory measurements. The PSV of the Earth's magnetic field in the central Pacific Ocean has been of marked interest, because the region was apparently characterized by a low non-dipole component of the geomagnetic field (Cox, 1962; Cox and Doell, 1964). The anomalously low secular variation in the region has persisted possibly for most of the last half million years, or even since 5.6 Ma (Cox and Doell, 1964; Doell and Cox, 1965, 1971, 1972; Cox, 1975). These studies have been instrumental in developing an understanding of the geomagnetic field in terms of spatial and temporal variations and in testing models of the palaeosecular variation and dynamo processes. The spatial and temporal PSV may, in turn, allow us to infer the topography and characteristics of the core–mantle interface and lateral uniformity in the lower mantle (e.g. Doell and Cox, 1972). Palaeosecular variation parameters are also important in studies of the time-averaged field and in testing the axial-dipole hypothesis.

A simple way to measure the PSV is to calculate the angular dispersion of palaeomagnetic data. The amount of angular dispersion measured about some assumed (or mean) direction or pole is important for a given locality, because different models of combinations of dipole and non-dipole

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components predict charateristic variations of this quantity with geographic latitude (Merrill and McElhinny, 1983; McFadden et al., 1988).

Several palaeomagnetic studies have been published for the Trans-Mexican Volcanic Belt (TMVB), but thus far only one study concerns the PSV (Herrero-Bervera et al., 1986). The non-dipole field intensity, as calculated from the International Geomagnetic Reference Field (IGRF), is rather low in the western and central part of Mexico and compares with that in the Central Pacific (e.g. Creer, 1963; Doell and Cox, 1972). It is possible that the anomalously low non-dipolar field intensity seen in the PSV for the Pacific region ('dipole window') could have extended up to the TMVB. Furthermore, the palaeomagnetic results of the lavas from the Marianas in the Western Pacific Ocean (U.S.–Japan Paleomagnetic Cooperation Program in Micronesia, 1975) have indicated that there may have been a belt of low average angular dispersion encircling the Earth between (roughly) 0 and 40° N. This belt would therefore include the TMVB, situated between about 18 and 20° N (Fig. 1).

Some of the data indicating anomalously low PSV at low latitudes from the Pacific are probably affected by insufficient and times-biased sampling. Nevertheless, recent extensive studies on lavas from Hawaii (McWilliams et al., 1982; Holcombe et al., 1986) support a low angular dispersion of the magnetic field at this locality for at least the last 40 000 years. Thus, it is important to pursue a more detailed data analysis of the palaeosecular variation from low latitudes, such as from the TMVB and central Mexico, to find the extent of the dipole window.

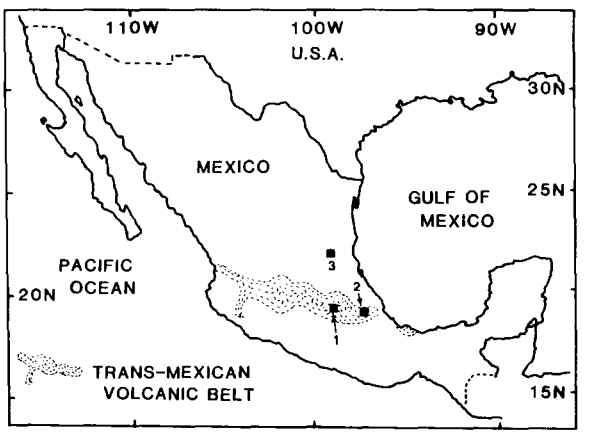


Fig. 1. Location map showing the three sampling localities in the area of study. 1, Chichinautzin Formation; 2, Eastern Trans-Mexican Volcanic Belt; 3, Stalagmite DAS2.

2. The data sets

The first data set for our PSV analysis is based on the radiocarbon and K–Ar dates published for the Chichinautzin Formation (Mooser et al., 1974; Bloomfield, 1975) just south of Mexico City. Previous geological and palaeomagnetic studies (Mooser et al., 1974; Herrero and Pal, 1977; Martin del Pozo, 1982; Herrero-Bervera et al., 1986) indicate that the age of the Chichinautzin Formation is $< 700\,000$ years, extending to historic times: the radiometric age of the Pedregal lava given by Libby (1965) is 2422 ± 250 years.

South of Mexico City, monogenetic volcanoes have been studied by Martin del Pozzo (1982), who extended the cone–flow morphology established by Bloomfield (1975) for the western part of the region to the region around the Chichinautzin volcano; about 150 cones and lava flows erupted within the last 40 000 years, although this time range may extend farther back (Bloomfield, 1975; Nixon et al., 1987). The larger part of the rocks studied belong to this age range, and we suppose that the volcanoes sampled this age range almost randomly, as monogenetic volcanoes are singular and independent events. The remaining data are from the Iztaccihuatl stratovolcano, which developed during the last 900 000 years. The studied lava flows probably have ages between around 80 000 and 580 000 years (Steele, 1971; Nixon et al., 1987). Steele (1971, 1985) points out that, as far as discernable, different lava flows were sampled, corresponding to 11 different vents with at least 26 flows. In conclusion we assume that the palaeomagnetic data used for the present study represent most of the Brunhes Chron, with about 70% of the sites corresponding to the last 40 000 years.

The palaeomagnetic data comprise 74 sites from Herrero-Bervera et al. (1986), Mooser et al. (1974),

TABLE 1

Palaeomagnetic data for Chichinautzin Formation volcanics [a]

Site number	Code	N	R	k	α_{95} (degrees)	I (degrees)	D (°E)	Latitude (°N)	Longitude (°E)	Age unit [c]
1	CHI-6	17	16.947	301	2.1	34.3	358.0	88.0	202.7	A [d]
2	CHI-8	9	8.843	51	7.3	27.1	7.2	83.1	2.6	A [d]
3	CHI-9	8	7.939	115	5.2	27.1	4.5	83.5	321.2	A [d]
4	CHI-10	9	8.932	118	4.8	34.9	6.0	84.3	10.4	A [d]
5	CHI-13	8	7.938	114	5.2	38.9	355.1	84.7	158.5	A [d]
6	CHI-15	7	6.903	62	7.7	33.6	356.1	86.2	202.3	A [d]
7 [b]	JU	7	6.979	289	3.5	22.4	323.2	53.8	164.5	A [e]
8	XA	5	4.984	242	4.9	22.1	343.8	72.6	146.9	A [e]
9	P-2	8	7.887	62	7.1	25.7	3.0	83.6	287.5	A [e]
10	MO-E-5	8	7.938	114	5.2	38.3	346.5	77.1	183.8	A [f]
11	MO-E-10	6	5.945	92	7.0	34.9	356.9	87.1	172.0	A [f]
12	MO-E-34	7	6.988	489	2.7	41.8	352.8	81.7	208.3	A [f]
13	CHI1/15	11	10.971	347	2.5	36.3	345.7	76.5	177.5	A [g]
14	CHI-1	8	7.904	73	6.5	26.7	357.6	84.5	253.9	B [d]
15	CHI-2	14	13.937	205	2.8	12.9	6.1	76.2	304.9	B [d]
16 [b]	CHI-3	12	11.981	577	1.8	46.6	330.3	61.5	165.4	B [d]
17	CHI-4	9	8.950	160	4.1	11.6	342.6	68.6	224.9	B [d]
18	CHI-5	11	10.915	118	4.2	18.6	345.5	72.9	221.4	B [d]
19	OZ	4	3.992	412	4.5	23.1	352.8	80.1	125.9	B [e]
20	ACO	7	6.926	82	6.7	32.6	357.6	87.3	138.5	B [e]
21	MO-E-28	7	6.976	253	3.8	33.4	358.5	88.2	132.0	B [f]
22	CHI-11	12	11.952	230	2.9	36.1	16.1	74.8	14.6	D [d]
23	TEU-1	7	6.969	200	4.2	17.1	338.7	66.9	147.2	E [e]
24	TEU-2	5	4.958	95	7.8	22.4	358.2	82.3	93.8	E [e]
25	MO-E-8	6	5.977	219	4.5	23.9	352.0	79.7	130.0	E [f]
26	MO-E-36	7	6.941	101	6.0	36.1	346.6	77.4	176.9	E [f]
27	MO-C-24	3	2.990	192	8.9	21.4	7.8	78.9	36.8	A–E [f]
28	MO-C-55	7	6.920	76	7.0	18.0	352.7	77.8	117.1	A–E [f]
29	MO-C-56	7	6.960	153	4.9	35.5	358.3	88.3	186.0	A–E [f]
30	MO-E-1	8	7.941	118	5.1	24.1	353.3	80.6	125.3	A–E [f]
31	MO-E-4	5	4.949	67	9.4	21.5	23.8	65.6	7.1	A–E [f]
32	MO-E-7	9	8.980	39	8.3	23.7	353.3	80.5	124.3	A–E [f]
33	MO-E-9	7	6.830	36	10.3	17.3	10.0	75.7	36.9	A–E [f]
34	MO-E-11	7	6.728	22	13.1	27.3	341.8	71.9	157.8	A–E [f]
35	MO-E-12	4	3.950	58	12.2	17.4	9.3	76.2	38.9	A–E [f]
36	MO-E-13	7	6.750	24	12.6	17.3	355.7	78.7	102.9	A–E [f]
37	MO-E-14	7	6.942	104	6.0	24.7	6.7	80.9	34.9	A–E [f]
38	MO-E-15	8	7.980	429	2.7	38.8	354.9	84.6	199.3	A–E [f]
39	MO-E-23	8	7.973	259	3.5	38.5	1.0	87.5	282.9	A–E [f]
40	MO-E-25	5	4.958	95	7.9	15.4	352.6	76.4	113.7	A–E [f]
41	MO-E-27	10	9.655	26	9.6	17.6	1.9	79.5	70.5	A–E [f]
42	MO-E-30	5	4.920	50	11.0	6.1	14.3	68.5	38.5	A–E [f]
43	MO-E-31	8	7.810	37	9.2	25.3	16.2	73.3	9.8	A–E [f]
44	MO-E-32	7	6.756	25	12.4	19.1	2.0	80.2	69.1	A–E [f]
45	MO-E-37	8	7.780	31	10.1	47.8	338.4	68.2	200.7	A–E [f]
46 [b]	MO-E-38	6	5.920	59	8.8	12.2	46.4	43.0	0.8	A–E [f]
47 [b]	MO-E-39	6	5.770	22	16.5	−7.9	37.8	46.1	18.9	A–E [f]
48	MO-E-41	7	6.902	61	7.8	32.8	348.9	79.4	164.3	A–E [f]
49	MO-E-43	6	5.847	33	11.9	35.2	2.8	87.4	349.7	A–E [f]
50	MO-E-44	8	7.909	77	6.4	35.1	9.7	80.9	349.4	A–E [f]
51	MO-E-45	5	4.839	24	15.6	18.3	5.9	78.5	50.3	A–E [f]
52	CHI-12	13	12.431	21	9.2	15.9	6.4	77.4	309.6	A–E [d]
53	IZT-29	7	6.917	73	7.1	38.1	7.4	82.7	332.4	F [h]

TABLE 1 (continued)

Site number	Code	N	R	k	α_{95} (degrees)	I (degrees)	D (°E)	Latitude (°N)	Longitude (°E)	Age unit [c]
54	IZT-30	7	6.924	79	6.8	31.9	356.6	86.2	141.2	F [h]
55	IZT-31	7	6.907	64	7.6	35.9	3.8	86.4	339.7	F [h]
56	IZT-32	7	6.861	43	9.3	30.0	356.7	85.6	127.2	F [h]
57	IZT-10	6	5.975	199	4.8	28.5	7.4	81.9	19.7	G [h]
58	IZT-11	7	6.964	168	4.7	23.5	3.6	82.2	54.2	G [h]
59	IZT-13	7	6.989	569	2.5	31.5	355.9	85.5	142.9	G [h]
60	IZT-16	7	6.808	31	11.0	38.6	13.4	77.2	337.5	G [h]
61	IZT-18	7	6.913	69	7.5	52.9	17.3	69.0	305.1	G [h]
62	IZT-19	7	6.811	68	10.9	51.0	20.6	67.7	313.3	G [h]
63 [b]	IZT-20	7	6.962	158	4.8	60.8	22.7	60.3	296.8	G [h]
64	IZT-21	7	6.954	132	5.3	32.1	352.4	82.6	161.4	G [h]
65	IZT-23	7	6.863	44	9.2	24.6	0.3	83.7	78.6	G [h]
66	IZT-24	7	6.926	82	6.7	21.3	355.3	80.7	111.0	G [h]
67 [b]	IZT-25	6	5.985	328	3.7	61.5	27.0	57.3	299.5	G [h]
68	IZT-78	7	6.975	243	3.9	28.3	1.3	85.7	64.4	G [h]
69	IZT-79	4	3.969	96	9.4	31.8	13.8	76.7	357.7	G [h]
70	IZT-82	8	7.961	178	4.2	39.8	356.1	85.0	215.1	G [h]
71	IZT-84	7	6.915	71	7.2	−8.0	355.5	66.4	92.6	G [h]
72	IZT-26	7	6.795	29	11.3	24.4	331.9	62.3	162.1	H [h]
73	IZT-27	6	5.937	80	7.6	30.0	354.0	83.5	143.8	H [h]
74	IZT-133	6	5.976	208	4.7	25.5	10.2	78.6	20.4	H [h]

[a] N, number of samples; R, vector sum; k and α_{95}, precision parameters; I and D, site mean inclination and declination; latitude and longitude of site mean VGP.

[b] Outlying sites.

[c] Age-units A–E are from geomorphological studies and supported by the following ^{14}C data: A, $\leqslant$ 8440 years BP; B, $\geqslant$ 19530 and 21860 years BP; D, $\geqslant$ 30200 and 30500 years BP; E, > 38590 years BP. Where the site location is not known precisely, A–E is given as age. F–H are age units according to Steele (1971) and Nixon (1987), who reports some K–Ar data: F, 80000 years BP; G, 270000, 340000 and 410000 years BP; H, 580000 years BP.

[d] Herrero-Bervera and Pal (1977).

[e] Herrero-Bervera et al. (1986).

[f] Mooser et al. (1974).

[g] This work.

[h] Steele (1985).

and Steele (1985) (Table 1). The specimens studied by Mooser et al. (1974) and Herrero-Bervera et al. (1986) from Chichinautzin were demagnetized by alternating fields. They are characterized by high stability remanence directions that remain stable at fields above 8 to 20 mT, as shown by linear trajectories towards the origin in the Zijderveld (1967) diagrams (Fig. 2).

In order to document the magnetic mineralogy of the carriers of remanance ore petrology, scanning electron microscope (SEM) observations, isothermal remanence (IRM) and thermomagnetic measurements were completed on selected samples. The main carriers are of low coercivity, very likely corresponding to titanomagnetites or magnetite. In some cases a small percentage of the initial intensity remains after 90 or 100 mT, indicating the presence of higher coercivity grains. Ore microscope observations indicate that most samples are characterized by less oxidized titanomagnetites of oxidation classes 1 and 2 (e.g. Ade Hall et al., 1968; Haggerty, 1976), with homogeneous sharp well-formed crystals, or with well-shaped grains with coarse ilmenite lamellae along (111) planes and fine needles of spinel exsolved along (100) cubic faces. The grain sizes are mainly below 50 μm, with elongated grains and also finer grained minerals. In some sites, maghemite was also observed. The presence of long needles of titanomagnetite and ilmenite intergrowths may

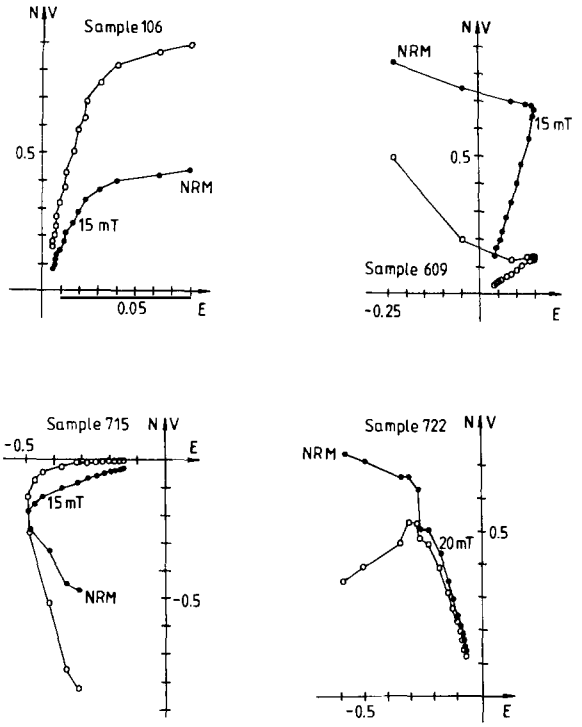


Fig. 2. Normalized a.f. demagnetization curves for samples lavas from the Chichinautzin Formation. Solid (open) symbols represent projections on the vertical (horizontal) plane.

account for the high magnetic stability observed. The IRM acquisition curves for most samples show saturation in fields < 0.1 T, supporting the occurrence of titanomagnetite and magnetite.

The second data set is from a 72-cm-high stalagmite from the cave Sotano del Arroyo near Valles, Mexico (22.09° N, 99.00° W) labelled as DAS2. Palaeomagnetic analyses have provided a record of secular variation from ~ 1200 years ago to the present. According to the authors (Latham et al., 1986), the age data of the stalagmite was provided by the U–Th method, and the concordancy was checked by ^{14}C and ^{226}Ra methods. The age of each one of the intervals is approximately 45 years.

The specimens cut from the stalagmite were a.f. demagnetized and exhibited stable remanences above ~ 10 mT. The NRM intensity of several specimens was extremely low, comparable to the noise level of the cryogenic magnetometer used. Therefore, the bulk of the specimens were demagnetized in 5.5 mT. Curie point determinations, IRM acquisition experiments, and ore microscope observations all point to magnetite as remanence carrier.

The third set of data is given by Böhnel (1985) and will be discussed in detail elsewhere. The study region is situated in the eastern part of the TMVB. According to Böhnel and Negendank (1981) and Negendank et al. (1982, 1985), the Altiplano area is characterized by basaltic, andesitic and rhyolitic volcanism, stratovolcanoes and domes rising above Cretaceous, NW–SE folded mountain ranges, and numerous monogenetic cones and maars mostly lying inside flat undrained basins. The igneous rocks concerned in the present study are of Quaternary age, but sites selected for the PSV study are limited to the Brunhes Chron. The radiocarbon, K–Ar, and pollen ages compiled by Negendank et al. (1985) show that the volcanic activity spans at least the last 100 000 years. This age range is several times longer than the maximum estimate of time (10 000 years) needed to average secular variation at a site (e.g. Creer, 1962a; Cox and Doell, 1964; Doell, 1969; Doell and Cox, 1971, 1972; McElhinny and Merrill, 1975; McWilliams et al., 1982; Lund, 1985).

The specimens were a.f. demagnetized, showing generally stable remanence directions (Fig. 3). Detailed studies of rock magnetism and magnetic mineralogy show that the magnetization is carried by low coercivity minerals, as of the titanomagnetite series. Curie temperatures range from 479 to 575° C. The median destructive fields (MDF) range from 20 to 70 mT and the Königsberger coefficient, Q, ranges from 4 to 343, indicating the presence of PSD and SD particles, and is in agreement with microscope observations. Ore mineralogy and SEM observations also show the predominance of titanomagnetites with low to intermediate degrees of oxidation. Further data and discussion will be presented elsewhere.

The virtual geomagnetic poles (VGP) are shown in Fig. 4 along with the mean positions and A_{95} confidence circles for each of the sampling localities.

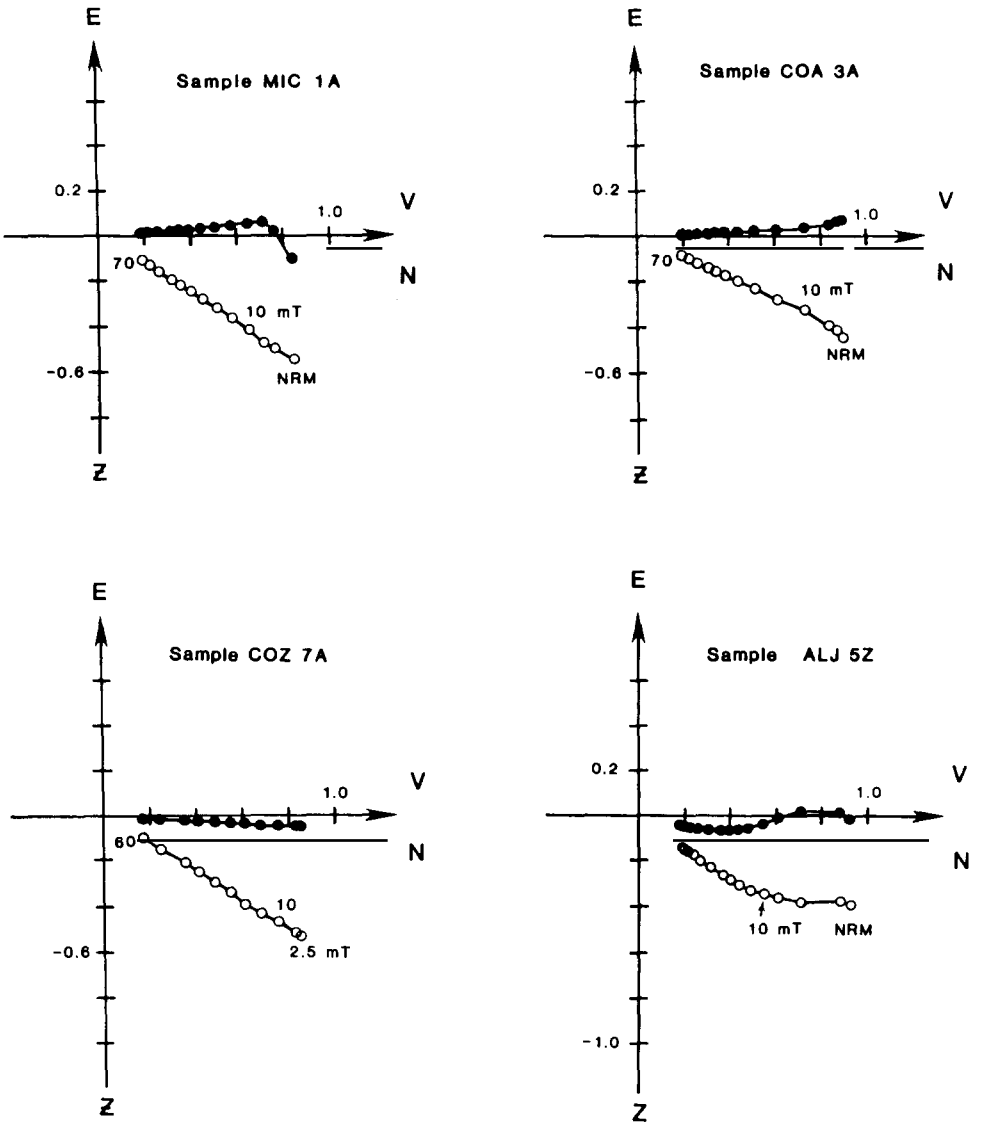


Fig. 3. Normalized a.f. demagnetization curves for samples from the lavas of the eastern part of the Trans-Mexican Volcanic Belt. Solid (open) symbols represent projections on the vertical (horizontal) plane.

3. Data analysis

When dealing with PSV it is common to use pole positions rather than directions (see discussion in McFadden et al., 1988). The dispersion of a data set is characterized by the standard deviation S or the precision parameter K (Fisher, 1953). Several sources contribute to the overall dispersion given by the parameters S_T and K_T. The most important are the within-site dispersion (S_W, K_W) and the between-site dispersion (S_B, K_B) (Watson and Irving, 1957). Smaller contributions are due to regional magnetic anomalies (S_A, K_A) (Doell and Cox, 1963). The within-site dispersion can be eliminated using the two-tier methods proposed by Watson and Irving (1957), Cox (1969), and modified by McFadden (1982), while the effect of regional magnetic anomalies is assessed from Doell and Cox (1963). Eliminating these latter contributions finally produces the dispersion

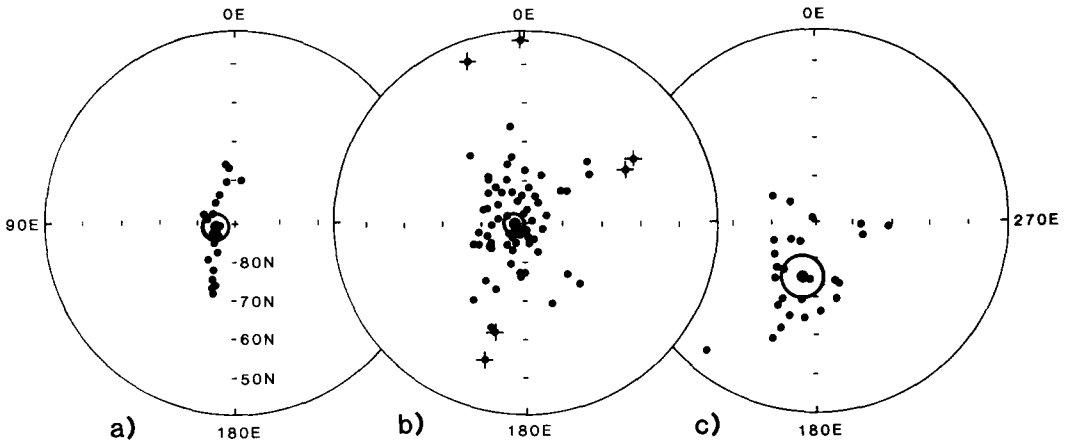


Fig. 4. Site mean virtual geomagnetic pole positions (small dots), palaeopoles (large dots) and confidence limits for three different localities: (a) Stalagmite DAS2, (b) Chichinautzin Formation (outlying VGPs are indicated by dots with a cross), (c) Eastern Trans-Mexican Volcanic Belt.

parameters S_F and K_F, which are believed to represent the PSV of the geomagnetic field. DAS2 represents a single site, and therefore does not require this treatment. As indicators for the shape of VGP distribution, the eccentricity E and axis ratio A/B (A and B define the major and minor axes of an ellipse approximated to the VGP distribution) may be determined (Engebretson and Beck, 1978). The larger E and A/B, the larger the deviation from a random distribution, which is here assumed to be Fisherian.

Alternatively, a shape analysis of the VGP distribution may be performed (Baag and Helsley, 1974). This was done only for the Chichinautzin data set. The observed radial distribution of the VGPs was compared by means of the chi-square test with a Fisher distribution. To calculate the azimuthal distribution, we counted the occurrence of VGPs in 10° longitude intervals and then calculated a 30° running mean to reduce a bias of the chi-square test (for random azimuthal distribution) due to the segment width and chosen start longitude. The chi-square test was then performed using 30° longitude intervals.

If any subset of data was suspected to represent the same volcanic event, all but one site (for ease, the one with the greatest number of samples) were excluded from the analysis. In this way, data from sites close together and showing indiscernible site mean directions were eliminated. To eliminate data that are not part of a random distribution (Fisher, 1953) at the 95% probability level we applied the criteria given by McFadden (1980). A non-random VGP distribution may be due to tectonic effects or to the occurrence of excursions/events of the geomagnetic field. Virtual geomagnetic poles affected by such effects have to be excluded from the PSV analysis (e.g. McFadden et al., 1988). On the other hand, eccentrically situated sources of the non-dipole field, e.g. at the core–mantle boundary, may produce a non-spherical VGP-pattern (Creer, 1983). Such a pattern could be transformed to a random distribution by weighting each VGP with the corresponding intensity of the palaeofield. This procedure could not be applied in the present study as palaeointensity determinations were not available.

4. Interpretation

From Fig. 4 and Table 2 it is easily seen that two of the three mean VGPs differ widely from the north geographic pole (NGP). We shall first discuss the case of the DAS2 stalagmite. According to the age determinations, the samples of this locality are nearly equally spaced in time. Assuming that each sample acquired its remanence at the time indicated by the radiometric ages, the directions and VGPs represent a time series with an

TABLE 2

Mean virtual geomagnetic poles and related dispersion parameters from central Mexico [a]

Locality [b]	B	K	A_{95} (degrees)	Latitude (°N)	Longitude (°E)	N	S_T (degrees)	S_W (degrees)	S_F (degrees)	E	A/B	Radial		Azimuthal	
												$\chi^2_{95\%}$	χ^2_{obs}	$\chi^2_{95\%}$	χ^2_{obs}
Chichinautzin (age range A–E)	47	48.4	3.0	87.3	122.6	7.7	11.7	9.8	11.1	0.70	1.39	14.1	9.2	16.9	3.4
Chichinautzin (age range F–G)	20	40.8	5.2	86.4	358.2	6.7	12.8	9.2	12.2	0.90	2.25	14.1	7.4	16.9	11.3
Chichinautzin (all data)	74	26.6	3.3	88.3	72.4	7.4	15.9	9.7	15.4	0.76	1.53	14.1	31.9	16.9	13.8
Chichinautzin (reduced data set)	68	43.7	2.6	87.9	95.1	7.4	12.3	9.8	11.7	0.68	1.36	14.1	12.1	16.9	7.6
Stalagmite DAS2	26	69.7	3.4	85.1	103.5	8.4	9.7	6.6	9.5	0.98	5.17				
Eastern TMVB	27	29.2	5.2	75.7	168.8	8.3	15.2	9.1	14.8	0.66	1.33				

[a] B, number of sites; K and A_{95}, precision parameters; latitude and longitude of mean VGP; N, mean number of samples per site; S_T, S_W, and S_F, angular dispersion parameters; E, eccentricity; A/B, ratio of major to minor axis of ellipse approximating the VGP distribution: $\chi_{95\%}$ and χ_{obs}, values for the chi-square test.

[b] For all localities but the Chichinautzin all data set calculations were made without outlying VGPs. Age ranges refer to Table 1.

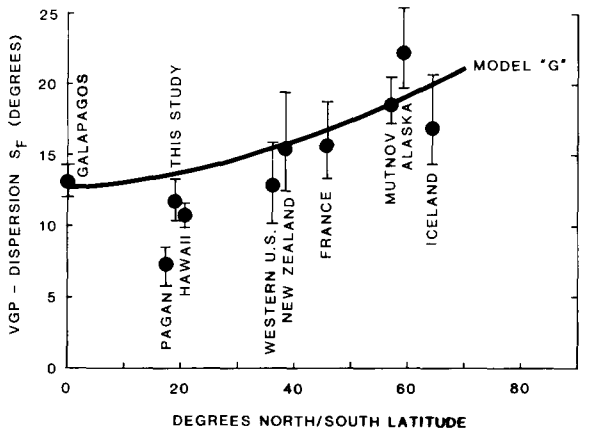


Fig. 5. Angular standard deviation for secular variation from various palaeomagnetic data sets (references: see Lund, 1985). The solid line shows the fit of model 'G' (McFadden et al., 1988) to the 5 Ma average results.

approximately 45-year sampling interval. The VGP distribution shown in Fig. 4a therefore may give the secular variation during the last ~ 1200 years in an almost continuous manner. The VGP distribution is strongly non-circular, also indicated by highly eccentric E and A/B values (Table 2). Obviously the time represented by these data was far from sufficient to allow secular variation to be averaged out. In terms of PSV the dispersion of this data set should therefore be lower than the dispersion of the palaeofield when averaged over a longer time interval. Indeed, $S_F = 9.5°$, much lower than the expected value of ~ 14° according, for example, to model G (McFadden et al., 1988) for the PSV (see Fig. 5).

The mean VGP for the eastern TMVB is left-tilted some 15° with respect to the NGP (Fig. 4c). As discussed by Böhnel (1985) and by Urrutia Fucugauchi and Böhnel (1987, 1988), this is probably due to counter-clockwise rotation of crustal blocks by neotectonic processes rather than to insufficient time coverage, since this data set comprises units with ages from historic times to more than 0.1 Ma. We may assume that the tectonic activity that resulted in this aberrant pole position was not a short-lived singular event. It is much more probable that the older volcanoes were tectonically affected to a larger extent than the younger volcanoes. Therefore, we expect that some VGPs may be unaffected, but others are left-tilted to different extents. Consequently, the observed VGP dispersion for this locality is composed by the secular variation of the palaeofield as well as by an unknown contribution due to neotectonic processes. Surprisingly, the observed VGP dispersion is $S_F = 14.8°$, which corresponds well to the expected value according to model G (Fig. 5) and values for E and A/B are low and do not reflect a 'smearing' of the VGPs orthogonal to the site longitude. Such a smearing could be observable in a much larger data set. On the assumption that the sampling of the Earth's magnetic field was adequate to average out secular variation and taking account of the additional tectonic contribution to the VGP dispersion in the eastern TMVB, the observed S_F is probably larger than the PSV in central Mexico during the upper Brunhes.

The third data set (Chichinautzin Formation) defines a mean pole near the NGP, which, in fact, is situated within the A_{95} confidence circle (Fig. 4b). This holds for the data set including all sites as well as the set in which six outliers are excluded. From the complete data set of 74 sites, a value of $S_F = 15.4°$ is obtained, which is very similar to the value from the eastern TMVB. Eliminating the six outliers by using the criteria of McFadden (1980), this value diminishes markedly to $S_F = 11.7°$, and is between the dispersion of the DAS2 stalagmite and the eastern TMVB. As these localities probably define unrealistic low and high PSV for central Mexico, the Chichinautzin data possibly represent the real secular variation of the palaeofield. Such an interpretation is supported by the large number of sites included in the analysis, and the sufficient wide age range apparently covered by the volcanic units.

Compared with theoretical and observed values for the PSV during the last 5 Ma, this value of S_F is much lower than expected, but very similar to the dispersion found in Hawaii (Fig. 5). The S_F also is lower than that previously reported by Herrero-Bervera et al. (1986) for the Chichinautzin formation. This is not because of the larger number of sites used in the present study, but because of the elimination of outliers. Applying the outlier criteria of McFadden (1980) to the data examined by Herrero-Bervera et al. (1986) results

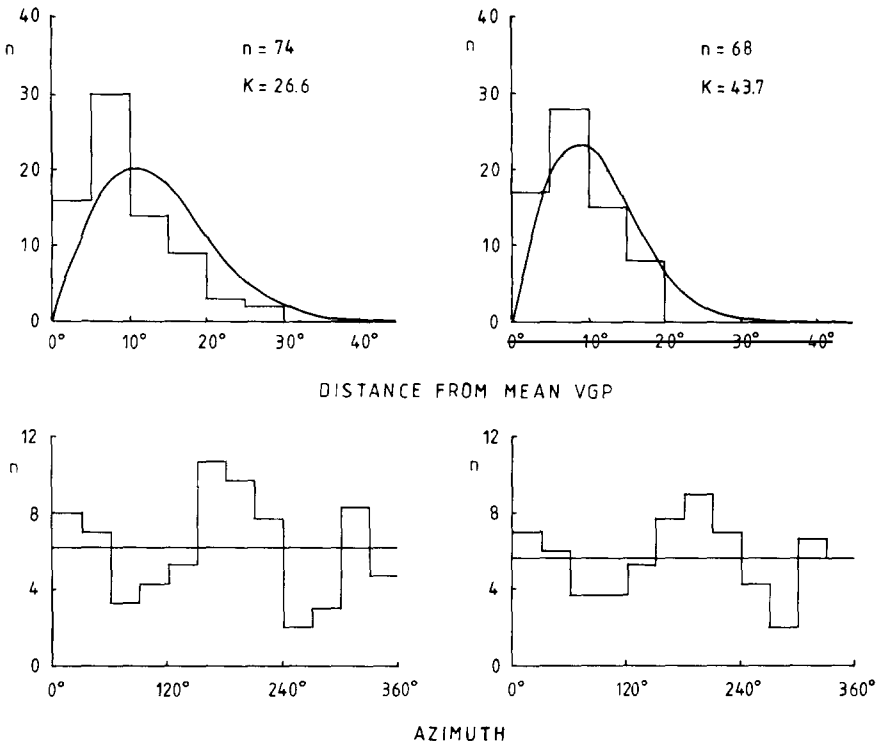


Fig. 6. Comparison of observed (step curves) and expected (continuous) VGP distribution. The left part of the diagram corresponds to the complete data set, the right part to the data set without outlying VGPs (see text).

in the elimination of the same data as in the present study and gives a dispersion of $S_F = 9.8°$.

5. Discussion

As outlined above, the data from the DAS2 stalagmite and from the eastern TMVB probably represent unrealistic dispersion of VGPs. Therefore, we will focus on the validity of the Chichinautzin data and their possible meaning for the PSV of the Earth's magnetic field. The value for S_F is reduced from 15.4° to 11.7° upon the elimination of six outlying VGPs. The anomalous position of such VGPs may be related to the following effects. First, the corresponding sampling sites could have suffered local tectonic movements, e.g. by tilt or rotation of parts of the lava flows. Although tilting cannot be rejected completely, it appears to be improbable as this sampling locality is well studied and there is no other independent evidence of it. Second, the position of the anomalous VGPs could correspond to the origin of the respective volcanic events during excursions/events of the Earth's magnetic field. According to the age of the Chichinautzin volcanic units, almost any excursion/event during the Brunhes may be included in the data (Jacobs, 1984). The above possibility may be supported by the grouping of the six outlying VGPs, defining three pairs of closely spaced VGPs (see Fig. 4b). It has already been observed in several other cases that large parts of palaeomagnetic data from Brunhes-age volcanics are affected by excursions/events (e.g. Böhnel et al., 1987; Geissmann et al., 1988). As excursions/events do not represent the 'normal' secular variation characterized by a random VGP distribution, such data are to be excluded for a PSV analysis (McFadden et al., 1988), as well as tectonically affected data. The elimination of the outlying VGPs is also supported by the shape analysis, by comparing the observed data

with a random distribution as defined by Fisher (1953). Figure 6 and the chi-square test (Table 2) show that only the reduced data set approximates such a distribution.

The obtained values of $S_F = 11.7°$ is much lower than expected, which could be due to inadequate sampling of the Earth's magnetic field in the study area. Inadequate sampling may result from a volcanic activity in irregular pulses rather than eruptions distributed uniformly in time (e.g. Cox, 1975). In the case of the Chichinautzin volcanic field, where 74 volcanic units were studied, a large number of volcanoes should have erupted simultaneously to produce a notable bias of the VGP distribution. Volcanological studies do not support this (Martin del Pozo, 1982), but according to field observations several small groups of volcanoes may exist, with coeval volcanoes within each group, which consequently would represent a multiple sampling of the Earth's magnetic field. If the age of such multiple eruptions is independent and randomly dispersed over several tens of thousands of years, this would not change the between site dispersion and S_F by much. Only the number of independent sites would be overestimated and the confidence limits of S_F would be too small. The upper and lower confidence limits for S_F as calculated here are 10.4° and 13.3°, respectively. Assuming that the number of really independent sites is only 40, corresponding to ~ 59% of the whole population, these limits would increase only slightly to 10.1° and 13.8°. Therefore, we propose that the obtained value of $S_F = 11.7°$ is probably characteristic for the PSV in central Mexico during the Brunhes Chron.

The data set can be subdivided into sites with ages less than ~ 50 000 years and sites with ages greater than ~ 80 000 years. Table 2 also includes the analyses of the subsets. Parameter S_F is lower for the younger than for the older rocks, with values of 11.1° and 12.2°, respectively. This could indicate a decrease of the VGP dispersion during the Brunhes. As the difference is not large and the 95% confidence limits overlap, this hypothesis needs to be proved by future studies.

A compilation of data from 541 lavas from all over the world, but situated at latitudes similar to that of the Chichinautzin Formation, resulted in a dispersion of $S_F = 13.4°$, which corresponds well to values predicted by model G (McFadden et al., 1988). The much lower dispersion of the Chichinautzin data, on the other hand, is comparable with that determined for the Hawaii Islands and other regions in the Pacific Basin (Fig. 5). Palaeomagnetic data for Hawaii provide a reliable record of the Earth's magnetic field at this locality since at least 5.6 Ma, and were interpreted as being due to the presence of a rather small non-dipole field beneath the sampling site (Cox, 1975). Analysis of palaeomagnetic data have further shown that in the Pacific Ocean region, PSV of the Cenozoic to Quaternary field was lower than for the rest of the world (e.g. Bingham and Stone, 1972; Creer et al., 1973). As the present-day central Pacific magnetic anomaly, characterized by a low non-dipole contribution, extends into central Mexico, the palaeomagnetic data here may reflect a similar anomalous field behaviour as in Hawaii, with low VGP dispersion. It is of great interest to study this anomaly of the Earth's magnetic field, its extension and its variation in time. The Trans-Mexican Volcanic Belt with several monogenetic volcano fields represents a most suitable study region for that purpose.

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